

Project: How does the performance of a Greenpower car change during a race?

I am going to use fake data to analyse how the performance of a Greenpower car change during a race.

Main objectives:

- See what conclusions can be made about the efficiency of empess.
- Identify the source of what causes empess to slow down during race.
- Use the data to make the process of looking for solutions that allow empess to maintain speed for as long as possible during races easier.

Display Table:

```
GreenpowerRaceData_fake_ = readtable('Greenpower race data (fake).xlsx');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
GreenpowerRaceData_fake_
```

```
GreenpowerRaceData_fake_ = 15×7 table
```

	LapNumber	LapTime_secs_	Var3	BatteryVoltage_V_	BatteryCurrent_A_
1	1	220.4000	NaN	24	28
2	2	239.4000	NaN	NaN	NaN
3	3	260.8000	NaN	NaN	NaN
4	4	279.7000	NaN	NaN	NaN
5	5	300.5000	NaN	NaN	NaN
6	6	319.3000	NaN	NaN	NaN
7	7	340.2000	NaN	NaN	NaN
8	8	360.9000	NaN	NaN	NaN
9	9	379.6000	NaN	NaN	NaN
10	10	399.2000	NaN	NaN	NaN
11	11	420.3000	NaN	NaN	NaN
12	12	440.6000	NaN	NaN	NaN
13	13	459.5000	NaN	NaN	NaN
14	14	480.1000	NaN	NaN	NaN

⋮

Import data from table into variables:

```
lapNumber = GreenpowerRaceData_fake_ .LapNumber
```

```
lapNumber = 15×1
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
⋮  
⋮
```

```
lapTime = GreenpowerRaceData_fake_ .LapTime_secs_
```

```
lapTime = 15×1
```

```
220.4000  
239.4000  
260.8000  
279.7000  
300.5000  
319.3000  
340.2000  
360.9000  
379.6000  
399.2000  
420.3000  
440.6000  
459.5000  
480.1000  
499.5000  
⋮  
⋮
```

```
batteryVoltage = GreenpowerRaceData_fake_ .BatteryVoltage_V_
```

```
batteryVoltage = 15×1
```

```
24  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
NaN  
⋮  
⋮
```

```
batteryCurrent = GreenpowerRaceData_fake_.BatteryCurrent_A_
```

```
batteryCurrent = 15×1
    28
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    :
```

```
lapDistance = GreenpowerRaceData_fake_.TrackDistance_metres_
```

```
lapDistance = 15×1
    3200
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    :
```

```
totalRaceTime = GreenpowerRaceData_fake_.TotalRaceTime_mins_
```

```
totalRaceTime = 15×1
    90
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    NaN
    :
```

```
avgLapTime = mean(lapTime)
```

```
avgLapTime =  
360
```

```
fastestLap = min(lapTime)
```

```
fastestLap =  
220.4000
```

```
slowestLap = max(lapTime)
```

```
slowestLap =  
499.5000
```

```
diff = slowestLap - fastestLap

diff =
279.1000
```

[illegible]

```
eachLapSpeed = lapDistanceDivider./lapTime
```

```
eachLapSpeed = 15x1
```

```

14.5191
13.3668
12.2699
11.4408
10.6489
10.0219
9.4062
8.8667
8.4299
8.0160
7.6136
7.2628
6.9641
6.6653
6.4064
⋮

```

Calculate Average lap speed:

```
avgLapSpeed = lapDistance(1)/avgLapTime
```

```

avgLapSpeed =
8.8889

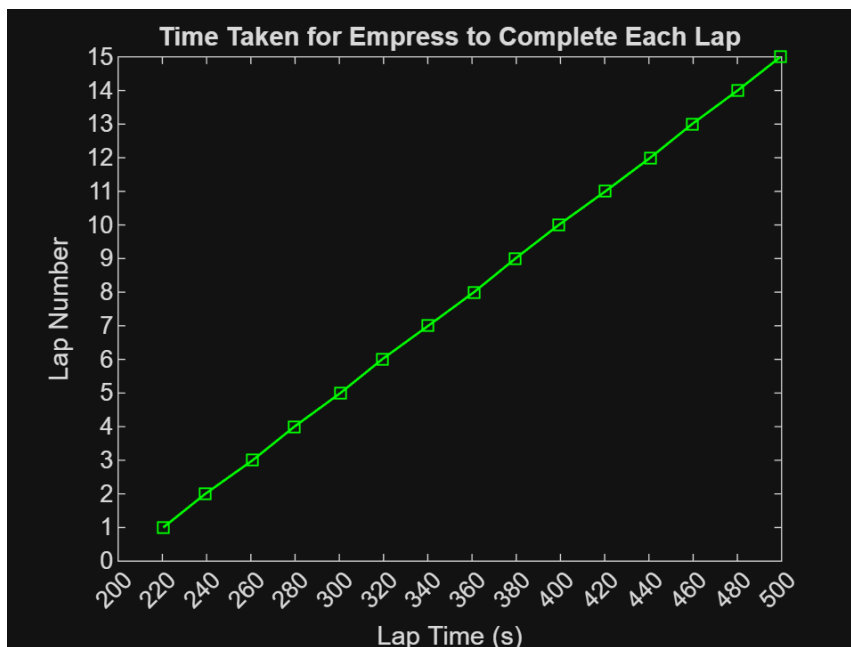
```

Graph to display the time taken for empres to complete each lap:

```

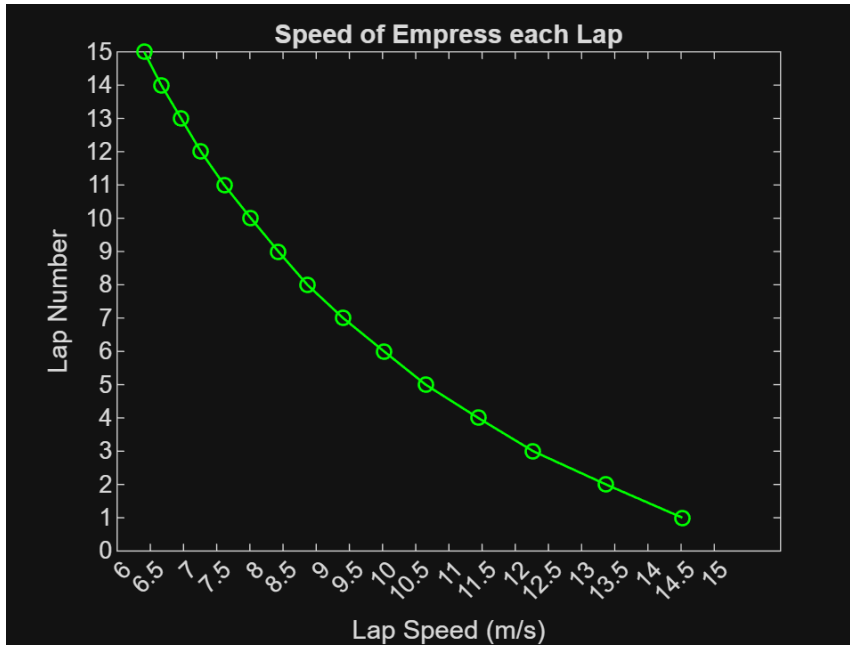
plot(lapTime,lapNumber,"gs-", LineWidth = 1)
xlabel("Lap Time (s)")
ylabel("Lap Number")
title("Time Taken for Empress to Complete Each Lap")
set(gca, "ytick", 0:1:15)
set(gca, "xtick", 0:20:500)

```



Graph to display the speed empress had each lap:

```
plot(eachLapSpeed, lapNumber, "go-", LineWidth = 1)
xlabel("Lap Speed (m/s)")
ylabel("Lap Number")
title("Speed of Empress each Lap")
set(gca, "xtick", 0:0.5:15)
set(gca, "ytick", 0:1:15)
```



Analysis of graphs:

Did performance improve or decline?

Empress' performance declined steadily.

Were the lap times consistent?

The strong positive correlation between lap time and lap number show the lap times were not consistent. Empress got slower and slower as the race went on.

At what point did performance change?

The speed of empress decreased quite steadily from the first lap. However, this is probably because the data is fake. In real races I have been to where empress fails to maintain initial speed, usually the time taken for the car to complete each lap is around the same for the first 30 minutes then gradually gets slower and slower until it's walking pace.

What might have caused the change?

- The battery started at low current
- While this can't be directly inferred from the graphs, we know that braking drains the car's battery.

Possible solutions?

- Use a battery with higher current e.g. 32A
- We could encourage drivers not to brake as much as possible.

Final conclusions from this project:

Benefits:

While it took longer to make calculations and draw graphs from data using MATLAB rather than excel, it allowed me to develop the coding skills I learnt from the MATLAB course I did in a practical and fun way. There are lots of parallels between coding in MATLAB and python, so it also helped me practice concepts in both coding languages. This is why I think it would be helpful for other students at Greenpower to explore.

Furthermore, MATLAB is a platform that is used to allow engineers and data analysts to make calculations and visualise data, which are two career paths I am highly interested in and now know I would have fun having either one of these careers because of the MATLAB course and this project.

In terms of answering my initial question (How does the performance of a Greenpower car change during a race?), MATLAB certainly helped me to do this. The graphs show a clear answer. More team collaboration will be essential to use this data to look for solutions that allow empress to maintain speed for as long as possible.

Next steps to improve this project:

- Would like to add a third graph showing power consumption over a race but would appreciate guidance about how to do this because I'm not sure if it's possible to record the current and voltage of the batteries over a race.
- Look for more efficient ways to code. This should be achieved through reading through more of the MATLAB help centre documentation and practice.
- Collaborate with the team to get suggestions about how to improve empress' performance.

Export code and results as a pdf

```
export("greenpower_race_analyser.mlx")
```

```
ans =  
'/MATLAB Drive/greenpower_race_analyser.pdf'
```